AMERICAN SOCIETY FOR MICROBIOLOGY | Journal of Clinical Microbiology®

BACTERIOLOGY

Check for updates

Reduced *In Vitro* Susceptibility of *Streptococcus pyogenes* to β-Lactam Antibiotics Associated with Mutations in the *pbp2x* Gene Is Geographically Widespread

James M. Musser,[a,b,c] Stephen B. Beres,[a,b] Luchang Zhu,[a,b] Randall J. Olsen,[a,b,c] Jaana Vuopio,[d,e] Hanne-Leena Hyyryläinen,[f] Kirsi Gröndahl-Yli-Hannuksela,[d] Karl G. Kristinsson,[g,h] Jessica Darenberg,[i] Birgitta Henriques-Normark,[j,k] Steen Hoffmann,[l] Dominque A. Caugant,[m] Andrew J. Smith,[n,o] Diane S. J. Lindsay,[o] David M. Boragine,[p] Timothy Palzkill[q]

[a]Center for Molecular and Translational Human Infectious Diseases Research, Houston Methodist Research Institute and Houston Methodist Hospital, Houston, Texas, USA
[b]Clinical Microbiology Laboratory, Department of Pathology and Genomic Medicine, Houston Methodist Research Institute and Houston Methodist Hospital, Houston, Texas, USA
[c]Department of Pathology and Laboratory Medicine, Weill Cornell Medical College, New York, New York, USA
[d]Institute of Biomedicine, University of Turku, Turku, Finland
[e]Turku University Hospital, Department of Clinical Microbiology, Turku, Finland
[f]Department of Health Security, Finnish Institute for Health and Welfare, Helsinki, Finland
[g]Department of Clinical Microbiology, Landspitali University Hospital, Reykjavik, Iceland
[h]Faculty of Medicine, School of Health Sciences, University of Iceland, Reykjavik, Iceland
[i]Public Health Agency of Sweden, Solna, Sweden
[j]Department of Microbiology, Tumor and Cell Biology, Karolinska Institutet, Stockholm, Sweden
[k]Department of Laboratory Medicine, Division of Clinical Microbiology, Karolinska University Hospital, Stockholm, Sweden
[l]Neisseria and Streptococcus Reference Laboratory, Bacteria, Parasites & Fungi, Infectious Disease Preparedness, Statens Serum Institut, Copenhagen, Denmark
[m]Division for Infection Control and Environmental Health, Norwegian Institute of Public Health, Oslo, Norway
[n]College of Medical, Veterinary and Life Sciences, Glasgow Dental Hospital and School, University of Glasgow, Glasgow, Scotland
[o]Scottish Microbiology Reference Laboratory, Glasgow, Scotland
[p]Department of Biochemistry and Molecular Biology, Baylor College of Medicine, Houston, Texas, USA
[q]Department of Pharmacology and Chemical Biology, Baylor College of Medicine, Houston, Texas, USA

James M. Musser, Stephen B. Beres, Luchang Zhu, and Randall J. Olsen are equally contributing co-first authors. The order of the authors is by mutual agreement.

ABSTRACT Recently, two related *Streptococcus pyogenes* strains with reduced susceptibility to ampicillin, amoxicillin, and cefotaxime, antibiotics commonly used to treat *S. pyogenes* infections, were reported. The two strains had the same nonsynonymous (amino acid-substituting) mutation in the *pbp2x* gene, encoding penicillin-binding protein 2X (PBP2X). This concerning report led us to investigate our library of 7,025 genome sequences of type *emm1*, *emm28*, and *emm89 S. pyogenes* clinical strains recovered from intercontinental sources for mutations in *pbp2x*. We identified 137 strains that, combined, had 37 nonsynonymous mutations in 36 codons in *pbp2x*. Although to a lesser magnitude than the two previously published isolates, many of our strains had decreased susceptibility *in vitro* to multiple beta-lactam antibiotics. Many *pbp2x* mutations were found only in single strains, but 16 groups of two or more isolates of the same *emm* type had an identical amino acid replacement. Phylogenetic analysis showed that, with one exception, strains of the same *emm* type with the same amino acid replacement were clonally related by descent. This finding indicates that strains with some amino acid changes in PBP2X can successfully spread to new human hosts and cause invasive infections. Mapping of the amino acid changes onto a three-dimensional structure of the related *Streptococcus pneumoniae* PBP2X suggests that some substitutions are located in regions functionally important in related pathogenic bacterial species. Decreased beta-lactam suscep-

Citation Musser JM, Beres SB, Zhu L, Olsen RJ, Vuopio J, Hyyryläinen H-L, Gröndahl-Yli-Hannuksela K, Kristinsson KG, Darenberg J, Henriques-Normark B, Hoffmann S, Caugant DA, Smith AJ, Lindsay DSJ, Boragine DM, Palzkill T. 2020. Reduced *in vitro* susceptibility of *Streptococcus pyogenes* to β-lactam antibiotics associated with mutations in the *pbp2x* gene is geographically widespread. J Clin Microbiol 58:e01993-19. https://doi.org/10.1128/JCM.01993-19.

Editor Alexander J. McAdam, Boston Children's Hospital

Copyright © 2020 American Society for Microbiology. All Rights Reserved.

Address correspondence to James M. Musser, jmmusser@houstonmethodist.org.

Received 3 December 2019
Returned for modification 23 December 2019
Accepted 15 January 2020
Accepted manuscript posted online 29 January 2020
Published 25 March 2020

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

tibility is geographically widespread in strains of numerically common *emm* gene subtypes. Enhanced surveillance and further epidemiological and molecular genetic study of this potential emergent antimicrobial problem are warranted.

KEYWORDS population genomics, bioinformatics, antibiotic resistance, whole-genome sequencing, public health

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

Generations of microbiologists, physicians, and others with strong interests in infectious diseases have been taught that *Streptococcus pyogenes* (group A streptococcus [GAS]) is universally susceptible to beta-lactam antimicrobial agents (1). Although the molecular basis for this resilient phenotype is unknown, given the global disease burden of greater than 700 million cases annually (2), universal susceptibility to these agents has been fortunate. Recently, Vannice et al. (3) identified two clonally related and epidemiologically linked strains of rare type *emm43.4 S. pyogenes* that had 8-fold higher MICs for ampicillin and amoxicillin and 3-fold higher MICs for cefotaxime, indicating decreased susceptibility to these antibiotics. The two strains had an identical single nonsynonymous (amino acid-altering) mutation in the *pbp2x* gene, encoding penicillin-binding protein 2X (PBP2X). This mutation confers a threonine-to-lysine replacement at amino acid 553 (Thr553Lys), a polymorphism that was not found in susceptible strains of type *emm43.4*. The authors suggested that the Thr553Lys replacement may be a first step toward *S. pyogenes* evolving resistance to beta-lactam antibiotics. The genomes of these two strains were sequenced as part of an outbreak investigation being done by Public Health—Seattle & King County in collaboration with the Centers for Disease Control and Prevention (CDC). Standard analysis conducted by the CDC GAS genome sequencing method includes identification of features potentially contributing to antibiotic resistance, including PBP2X variants (4). Through this process, the *pbp2x* missense mutations associated with decreased antibiotic susceptibility reported by Vannice et al. (3) were detected. The identification of these two strains is concerning and may signal a substantial public health problem because beta-lactams remain the frontline treatment globally for the majority of GAS infections.

To assess the potentially unrecognized broader extent of this inauspicious discovery, we felt compelled to interrogate our library of 7,025 genome sequences of type *emm1*, *emm28*, and *emm89 S. pyogenes* clinical isolates from intercontinental sources for nonsynonymous mutations in *pbp2x*. Bioinformatic analysis identified 137 strains with 37 amino acid changes at 36 sites in *pbp2x* that could alter MIC values for beta-lactam antibiotics. A subset of strains with *pbp2x* mutations was analyzed for beta-lactam MIC values using the gradient method (Etest strips). Our results indicate that clinical isolates with *pbp2x* mutations associated with small decreases in beta-lactam susceptibility in this common human-specific pathogen are more widespread than appreciated. Enhanced surveillance and fuller epidemiological and molecular genetic study of this potentially emergent antimicrobial problem are warranted.

MATERIALS AND METHODS

***S. pyogenes* strains and whole-genome sequence data.** The *emm1* ($n = 3,615$), *emm28* ($n = 2,095$), and *emm89* ($n = 1,315$) *S. pyogenes* strains that we studied have been described in our previous publications (5–8). The genome sequence data generated with Illumina instruments have previously been deposited in publicly available databases in the National Center for Biotechnology Information Sequence Read Archive (BioProject accession numbers PRJNA236767, PRJNA434389, PRJNA287922, and PRJNA387243). Nucleotide polymorphisms in the *pbp2x* gene in these strains were identified by bioinformatics methods that have been extensively described previously (5).

Antibiotic susceptibility determinations. Forty-two strains with nonsynonymous mutations in *pbp2x* were tested for potential decreased susceptibility to penicillin by plating on tryptic soy agar supplemented with 6-ng/ml penicillin G (benzylpenicillin) or 15-ng/ml ampicillin (9, 10). These strains represent a diverse array of organisms with distinct *pbp2x* mutations from *emm1*, *emm28*, and *emm89* organisms. Six reference strains lacking *pbp2x* mutations (wild-type [WT] strains, consisting of one strain of *emm1*, three strains of *emm28*, and two strains of *emm89*) were included as PBP2X consensus wild-type comparators. The reference strains are *emm* type and genetic clade matched and have the most common allele representative of their genetic background for global transcriptional regulators of known virulence factors, and several have been extensively studied both *in vitro* and in animal virulence experiments. The

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

plates were incubated overnight at 37°C in 5% CO_2, and growth was assessed as present or absent. MIC values for six beta-lactam antibiotics (ampicillin, penicillin G, cefotaxime, cefoxitin, ceftazidime, and meropenem) were determined by the gradient method (Etest strips; bioMérieux) using standard clinical laboratory procedures. MIC values were scored independently by three investigators. Some strains were also tested for penicillin G and ampicillin susceptibility by broth microdilution in Todd-Hewitt broth supplemented with 0.5% yeast extract (THY). Liquid cultures were incubated overnight at 37°C in 5% CO_2, and growth was determined by determination of the optical density at 600 nm.

Phylogenetic analysis of whole-genome sequence data. The phylogeny among the strains was inferred by neighbor joining based on concatenated sequential core chromosomal single nucleotide polymorphisms (SNPs) by methods described previously (5). To constrain inferences to predominantly vertically acquired SNPs, regions of recombination were predicted based on entire core genome sequences using the Gubbins algorithm, and putatively horizontally acquired SNPs were excluded. Clades of related strains were defined using hierarchical Bayesian analysis of population structure (hierBAPS), also as previously described (5).

Construction of isogenic strain MGAS27213-PBP2X-WT. Isogenic strain MGAS27213-PBP2X-WT was constructed by replacing the naturally occurring mutant *pbp2x* gene (Pro601Leu) of clinical isolate MGAS27213 with the wild-type allele encoding Pro601 using procedures previously described (11). Briefly, wild-type *pbp2x* of strain MGAS27566 was amplified by PCR using primers pbp2x-1 (5′-GTGAAT ACATGCGATAGGAGAACTCCAG-3′) and pbp2x-2 (5′-CAATTGTACATTGATTCGCCAACTAAGTC-3′). The PCR amplicon was cloned into suicide vector pBBL740 and then transformed into parental strain MGAS27213, encoding the mutant *pbp2x* allele (Pro601Leu). Whole-genome sequencing of isogenic strain MGAS27213-PBP2X-WT confirmed that the mutant *pbp2x* allele was replaced by the wild-type *pbp2x* allele and that the constructed strain lacked spurious mutations.

PBP2X structure modeling. The crystal structure of PBP2X from *Streptococcus pneumoniae* (PDB accession number 1RP5, chain A) was used to map the location of the amino acid substitutions relative to the active site of the transpeptidase domain. This structure was used because the structure of *S. pyogenes* PBP2X has not been determined. The two PBP2X proteins are well conserved in both amino acid sequence (54.1% identical, 82.1% similar) and structural fold, and PBP2X from *S. pneumoniae* has been well studied by several investigators (12–16). The *S. pyogenes* amino acid substitutions were mapped onto the *S. pneumoniae* PBP2X structure using the Chimera program (17). Chimera was also used to align PBP2X with PBP3 from *Pseudomonas aeruginosa* (PDB accession number 6UN3) and PBP2a from *Staphylococcus aureus* (PDB accession number 1VQQ, chain A) to assign the role of each residue in relation to PBP2X within *S. pyogenes*.

RESULTS

Identification of PBP2X amino acid replacements. To test the hypothesis that the *S. pyogenes* strains in our international collection of human clinical isolates contained polymorphisms in the *pbp2x* gene, we interrogated the population genomic data generated in our previous studies of *emm1*, *emm28*, and *emm89* organisms (5–8). The vast majority of these strains were recovered from a normally sterile site of patients with invasive infections, such as bacteremia and necrotizing fasciitis. Among the 7,025 whole-genome sequences examined, we identified 137 strains that in the aggregate had 37 nonsynonymous mutations altering 36 codons of the 2,259-nucleotide *pbp2x* gene. We also identified 161 strains with a synonymous single nucleotide polymorphism (that is, a silent mutation that would not alter the amino acid sequence of PBP2X) at 10 positions, each in a separate codon of *pbp2x*. Thus, 79% of SNP sites resulted in an amino acid replacement, a significantly greater percentage than expected by chance alone (for the 48 *pbp2x* alleles in the cohort by the Nei-Gojobori method, the ratio of rates of nonsynonymous/synonymous site substitutions [K_a/K_s] = 1.49; Fisher exact test, $P = 5.27e{-}41$). This elevated percentage of nonsynonymous mutations is consistent with the effect of positive selection acting on *pbp2x*. Among the strains with nonsynonymous mutations, with a single exception, each of the 137 strains had only one amino acid replacement relative to the consensus wild-type PBP2X sequence. The exception was an *emm28* strain (MGAS28532) recovered in the United States that had a unique combination of two contiguous amino acid replacements (Phe599Tyr and Gly600Asp) in PBP2X (Table 1). Of note, none of the 7,025 isolate sequences interrogated had an insertion or deletion mutation in *pbp2x*, indicating that the peptidoglycan transpeptidase function of PBP2X is essential. This finding is consistent with the results of saturating transposon mutagenesis screens, which, during library generation, also failed to recover strains with integrations in *pbp2x* (18, 19).

The analysis identified four sites that had the same amino acid replacement (Gly288Ser, Met342Ile, Gly600Asp, and Pro601Leu) present in multiple *emm* types (Fig. 1). In each case, these amino acid replacements were represented among strains of

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

TABLE 1 PBP2X amino acid replacement strains and PBP2X wild-type comparator strains[a]

Strain (MGAS no.)	*emm* type	Amino acid substitution	Country	Yr	Capsule	Results on agar with:		MIC (μg/ml)					
						Penicillin G at 6 ng/ml	Ampicillin at 15 ng/ml	Penicillin G[b]	Ampicillin	Cefotaxime	Cefoxitin	Ceftazidime	Meropenem
2221	1	Wild type	Australia	1998	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
23877	1	Wild type	Canada	2002	Positive			0.012*	0.016				
24791	1	Wild type	Finland	2006	Positive			0.012*	0.016				
7867	28	Wild type	Canada	1991	Negative			0.012*	0.016				
8357	28	Wild type	Finland	1996	Negative			0.012*	0.016				
10778	28	Wild type	Canada	1998	Negative			0.012*	0.016				
10783	28	Wild type	Canada	1998	Negative			0.012*	0.016				
11052	28	Wild type	Finland	2000	Negative			0.012*	0.016				
27961	28	Wild type	United States	2005	Negative	Negative	Negative	<0.016	0.016	0.023	1	0.19	0.006
28426	28	Wild type	United States	1999	Negative	Negative	Negative	<0.016	0.016	0.032	1.5	0.19	0.004
28737	28	Wild type	United States	2012	Negative	Negative	Negative	<0.016	0.016	0.032	1.5	0.19	0.003
28905	28	Wild type	United States	2004	Negative			0.012*	0.016				
23530	89	Wild type	Italy	1997	weak			0.012*	0.016				
26568	89	Wild type	United States	1996	Positive			0.012*	0.016				
26645	89	Wild type	United States	2009	Positive			0.012*	0.016				
26844	89	Wild type	United States	2008	Negative	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
27545	89	Wild type	Finland	2010	weak			0.012*	0.016				
27566	89	Wild type	Finland	2011	Negative	Negative	Negative	<0.016	0.016	0.032	1.5	0.19	0.006
29554	28	Ile47Val	Finland	2004	Negative	Negative	Negative	<0.016	<0.016	0.016	0.75	0.125	0.004
29632	28	Ile47Val	Finland	2002	Negative								
28315	28	Asp52Gly	United States	1998	Negative	Negative	Negative	<0.016	0.016	0.023	1	0.19	0.004
28329	28	Asp52Gly	United States	1998	Negative								
28433	28	Asp52Gly	United States	2003	Negative								
28894	28	Asp52Gly	United States	2004	Negative								
31966	28	Thr70Asn	Norway	2014	Negative								
5534	1	Gly85Ser	Finland	1995	Positive	Negative	Negative	<0.016	<0.016	0.016	1	0.19	0.006
5546	1	Gly85Ser	Finland	1995	Positive								
5555	1	Gly85Ser	Finland	1995	Positive								
5556	1	Gly85Ser	Finland	1995	Positive								
28692	28	Ser92Phe	United States	2008	Negative								
26637	89	Met171Ile	United States	2009	Negative								
26639	89	Met171Ile	United States	2009	Negative								
26667	89	Met171Ile	United States	2009	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.25	0.006
29128	28	Ala174Val	Finland	2014	Negative	Negative	Negative	<0.016	<0.016	0.012	0.75	0.125	0.003
29176	28	Ala174Val	Finland	2013	Negative								
29178	28	Ala174Val	Finland	2013	Negative								
29316	28	Ala174Val	Finland	2010	Negative								
27413	89	Asp233Asn	United States	2011	Negative								
29408	28	Thr245Ile	Finland	2008	Negative	Negative	Negative	<0.016	<0.016	0.012	0.75	0.125	0.003
29254	28	Phe274Leu	Finland	2012	Negative	Positive	Negative	0.016	0.023	0.023	2	0.19	0.006
28415	28	Val281Ile	United States	2003	Negative								
28772	28	Gly288Ser	United States	2012	Negative								
28773	28	Gly288Ser	United States	2012	Negative	Positive	Positive	0.016	0.023	0.032	1.5	0.25	0.006
26860	89	Gly288Ser	United States	2003	Positive	Positive	Positive	0.016	0.023	0.047	2	0.38	0.008

(Continued on next page)

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

TABLE 1 (Continued)

Strain (MGAS no.)	*emm* type	Amino acid substitution	Country	Yr	Capsule	Results on agar with:		MIC (μg/ml)					
						Penicillin G at 6 ng/ml	Ampicillin at 15 ng/ml	Penicillin G[b]	Ampicillin	Cefotaxime	Cefoxitin	Ceftazidime	Meropenem
26929	89	Gly288Ser	United States	2003	Positive	Positive	Positive	0.016	0.023	0.016	1.5	0.19	0.006
26932	89	Gly288Ser	United States	2003	Positive	Positive	Positive	0.016	0.023	0.032	1.5	0.25	0.006
27438	89	Gly288Ser	United States	2006	Positive	Positive	Positive	0.016	0.023	0.047	1.5	0.25	0.008
28711	28	Thr323Met	United States	2011	Negative								
10786	28	Met342Ile	Canada	1998	Negative	Positive	Positive	0.016	0.016	0.023	2	0.19	0.004
28367	28	Met342Ile	United States	2003	Negative								
27033	89	Met342Ile	United States	2008	Positive	Positive	Positive	0.023	0.023	0.032	3	0.25	0.006
7898	28	Lys422Arg	Canada	1995	Negative	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.004
7922	28	Lys422Arg	Canada	1996	Negative								
7973	28	Lys422Arg	Canada	1997	Negative								
10752	28	Lys422Arg	Canada	1992	Negative								
29061	28	Phe425Leu	Finland	2015	Negative	Positive	Positive	0.016	0.016	0.023	2	0.19	0.006
29068	28	Phe425Leu	Finland	2015	Negative								
29093	28	Phe425Leu	Finland	2014	Negative								
29125	28	Phe425Leu	Finland	2014	Negative								
29141	28	Phe425Leu	Finland	2014	Negative								
30068	28	Phe425Leu	Finland	2015	Negative								
7976	28	Ala438Val	Canada	1997	Negative	Positive	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
28782	28	Thr461Pro	United States	2012	Negative	Positive	Negative	0.016	0.016	0.016	1.5	0.125	0.006
26899	89	Gln462His	United States	2003	Negative	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
25070	1	Thr515Aal	Norway	2004	Positive	Negative	Negative	<0.016	<0.016	0.016	1	0.125	0.003
28088	28	Gly521Ser	United States	2010	Negative	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.004
27590	89	Pro526Leu	Finland	2011	Negative	Positive	Negative	<0.016	0.016	0.023	2	0.25	0.006
27612	89	Pro526Leu	Finland	2012	Negative	Positive	Negative	0.016	0.016	0.032	2	0.25	0.008
23930	1	Thr535Ile	Canada	2003	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
28532	28	Phe599Tyr and Gly600Asp	United States	2006	Negative	Positive	Positive	0.023	0.032	0.023	1.5	0.19	0.012
7873	28	Gly600Asp	Canada	1992	Negative								
7935	28	Gly600Asp	Canada	1996	Negative								
7936	28	Gly600Asp	Canada	1996	Negative								
7937	28	Gly600Asp	Canada	1996	Negative								
7938	28	Gly600Asp	Canada	1996	Negative								
7947	28	Gly600Asp	Canada	1996	Negative								
7948	28	Gly600Asp	Canada	1996	Negative								
7953	28	Gly600Asp	Canada	1996	Negative								
7987	28	Gly600Asp	Canada	1997	Negative								
7989	28	Gly600Asp	Canada	1997	Negative								
7994	28	Gly600Asp	Canada	1997	Negative								
8014	28	Gly600Asp	Canada	1999	Negative								
8015	28	Gly600Asp	Canada	1999	Negative								
8016	28	Gly600Asp	Canada	1999	Negative								
10813	28	Gly600Asp	Canada	2000	Negative								
27982	28	Gly600Asp	United States	2005	Negative								
28165	28	Gly600Asp	United States	1998	Negative								

(Continued on next page)

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

TABLE 1 (Continued)

Strain (MGAS no.)	*emm* type	Amino acid substitution	Country	Yr	Capsule	Results on agar with:		MIC (μg/ml)					
						Penicillin G at 6 ng/ml	Ampicillin at 15 ng/ml	Penicillin G[b]	Ampicillin	Cefotaxime	Cefoxitin	Ceftazidime	Meropenem
28336	28	Gly600Asp	United States	2000	Negative								
28380	28	Gly600Asp	United States	2003	Negative								
28425	28	Gly600Asp	United States	1999	Negative	Positive	Positive	0.016	0.023	0.032	1.5	0.19	0.006
28792	28	Gly600Asp	United States	2000	Negative	Positive	Positive	0.016	0.023	0.032	1.5	0.19	0.006
27143	89	Gly600Asp	United States	2012	Negative	Positive	Positive	0.016	0.023	0.032	1.5	0.125	0.006
27326	89	Gly600Asp	United States	2013	Negative	Positive	Positive	<0.016	0.016	0.023	1	0.094	0.004
29393	28	Pro601Leu	Finland	2008	Negative	Positive	Positive	0.023	0.047	0.047	1.5	0.25	0.012
26837	89	Pro601Leu	United States	2011	Negative								
27213	89	Pro601Leu	United States	2012	Negative	Positive	Positive	0.032	0.047	0.064	1.5	0.38	0.012
27308	89	Pro601Ser	United States	2013	Negative	Negative	Negative	<0.016	0.016	0.032	1.5	0.25	0.003
27316	89	Pro601Leu	United States	2013	Negative	Positive	Positive	0.032	0.047	0.064	1.5	0.38	0.016
27453	89	Pro601Leu	United States	2010	Negative	Positive	Positive	0.032	0.047	0.064	1.5	0.38	0.012
31677	89	Asp620Tyr	Scotland	2016	Negative	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
23875	1	Arg632His	Canada	2002	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
23888	1	Arg632His	Canada	2002	Positive								
23904	1	Arg632His	Canada	2002	Positive								
23905	1	Arg632His	Canada	2002	Positive								
23906	1	Arg632His	Canada	2002	Positive								
23950	1	Arg632His	Canada	2003	Positive								
23980	1	Arg632His	Canada	2003	Positive								
24056	1	Arg632His	Canada	2004	Positive								
24141	1	Arg632His	Canada	2002	Positive								
24177	1	Arg632His	Canada	2002	Positive								
25364	1	Arg632His	Sweden	1998	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
25387	1	Arg632His	Sweden	2000	Positive								
25388	1	Arg632His	Sweden	2000	Positive								
28323	28	Gly647Asp	United States	2001	Negative								
10792	28	Ser651Gly	Canada	1998	Negative								
26743	89	Val662Ile	United States	1999	Negative								
24008	1	Arg692Cys	Canada	2003	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
24019	1	Arg692Cys	Canada	2003	Positive								
24202	1	Arg692Cys	Canada	1999	Positive								
24203	1	Arg692Cys	Canada	1999	Positive								
24212	1	Arg692Cys	Canada	1999	Positive								
24645	1	Arg692Cys	Sweden	2009	Positive								
24692	1	Arg692Cys	Sweden	2009	Positive								
24709	1	Arg692Cys	Sweden	2009	Positive								
25599	1	Glu695Asp	United States	2010	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
25603	1	Glu695Asp	United States	2010	Positive								
1350	1	Lys708Arg	Germany[c]	1984	Positive	Negative	Negative	<0.016	0.016	0.023	1.5	0.19	0.006
27009	89	Lys730Arg	United States	2007	Negative								
23858	1	Asp734Gly	Iceland	2006	Positive	Negative	Negative	0.016	0.016	0.023	1.5	0.19	0.004
25205	1	Asp734Gly	Sweden	2005	Positive								
25221	1	Asp734Gly	Sweden	2007	Positive	Negative	Negative	0.016	0.023	0.023	1.5	0.25	0.006

(Continued on next page)

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

TABLE 1 (Continued)

Strain (MGAS no.)	*emm* type	Amino acid substitution	Country	Yr	Capsule	Results on agar with:		MIC (μg/ml)					
						Penicillin G at 6 ng/ml	Ampicillin at 15 ng/ml	Penicillin G[b]	Ampicillin	Cefotaxime	Cefoxitin	Ceftazidime	Meropenem
26024	1	Asp734Gly	Denmark	2002	Positive								
26052	1	Asp734Gly	Denmark	2002	Positive								
26055	1	Asp734Gly	Denmark	2002	Positive								
26056	1	Asp734Gly	Denmark	2002	Positive								
26057	1	Asp734Gly	Denmark	2002	Positive								
26059	1	Asp734Gly	Denmark	2002	Positive								
26060	1	Asp734Gly	Denmark	2002	Positive								
26064	1	Asp734Gly	Denmark	2003	Positive								
26071	1	Asp734Gly	Denmark	2003	Positive								
26082	1	Asp734Gly	Denmark	2003	Positive								
26083	1	Asp734Gly	Denmark	2003	Positive								
26085	1	Asp734Gly	Denmark	2003	Positive								
26096	1	Asp734Gly	Denmark	2003	Positive								
26101	1	Asp734Gly	Denmark	2003	Positive								
26109	1	Asp734Gly	Denmark	2003	Positive								
26113	1	Asp734Gly	Denmark	2004	Positive								
26159	1	Asp734Gly	Denmark	2009	Positive								
26197	1	Asp734Gly	Denmark	2005	Positive								
26213	1	Asp734Gly	Denmark	2006	Positive								
26231	1	Asp734Gly	Denmark	2006	Positive								
26267	1	Asp734Gly	Denmark	2007	Positive								
26271	1	Asp734Gly	Denmark	2008	Positive								
26279	1	Asp734Gly	Denmark	2008	Positive								
26290	1	Asp734Gly	Denmark	2008	Positive								

[a]Summary of available data for 137 *emm1*, *emm28*, and *emm89 S. pyogenes* strains with amino acid replacements in PBP2X and 18 PBP2X wild-type comparator strains.

[b]Penicillin G MIC assays were done using test strips with either of two dose ranges, consisting of the standard range of 0.016 to 256 μg/ml or a low range 0.002 to 32 μg/ml; assays done using the low-dose-range strips are indicated (*).

[c]German Democratic Republic, the former East Germany.

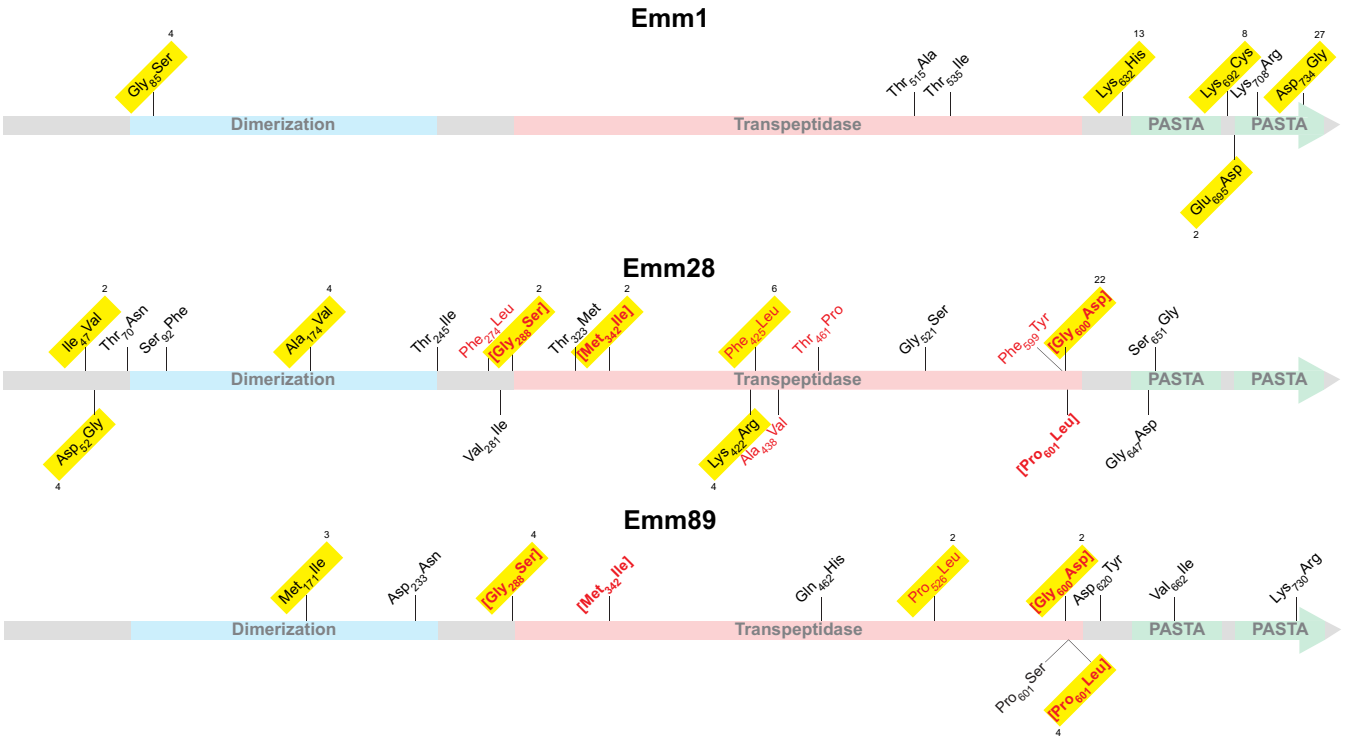


FIG 1 Location of PBP2X amino acid replacements identified among the 7,025 genomes of *emm1*, *emm28*, and *emm89* clinical isolates. Amino acid replacements identified in multiple strains are highlighted in yellow, with superscripts denoting the number of strains. Replacements identified in both *emm28* and *emm89* strains are in bold and enclosed in brackets. Replacements associated with reduced susceptibility to one or more of the beta-lactam antibiotics tested under the *in vitro* conditions analyzed are shown in red. The dimerization, transpeptidase, and PASTA domains are indicated.

types *emm28* and *emm89* (Fig. 1). Additionally, these four replacements were present among multiple isolates within a single *emm* type. The finding of the same replacements both in multiple *emm* types and in multiple isolates of the same *emm* type strongly suggests that these changes have been selected by exposure to beta-lactam antibiotics. In contrast, there was no example of sharing of amino acid replacements between type *emm1* strains and either *emm28* or *emm89* strains. Despite the *emm1* cohort comprising the greatest number of isolates ($n = 3{,}615$), it had a lower frequency of nonsynonymous SNP sites ($n = 8$) than either the *emm28* (2,095 isolates and 21 nonsynonymous sites) or *emm89* (1,315 isolates and 12 nonsynonymous sites) cohort. Moreover, the nonsynonymous SNP sites among the *emm1* isolates differed in distribution compared to their distribution in the *emm28* and *emm89* isolates, being among the *emm1* isolates somewhat less prevalent in the middle (i.e., the transpeptidase domain) and more prevalent toward the 3′ end of *pbp2x* (i.e., the penicillin-binding protein and serine/threonine kinase-associated [PASTA] domains).

An alignment of PBP2X of *S. pyogenes*, *S. pneumoniae*, and *Streptococcus agalactiae* shows that the Gly residue at position 288 and the Met residue at position 342 are conserved among the three species (Fig. 2). Of note, Met342 is located in the conserved SXXK motif, containing the transpeptidase activity catalytic Ser residue.

Phylogenetic analysis of strains of the same *emm* type with the identical PBP2X amino acid replacement using whole-genome sequence data. We identified 16 instances in which two or more strains of the same *emm* type had the identical amino acid replacement (Table 1 and Fig. 1). In general, strains of the same *emm* type with the identical *pbp2x* nonsynonymous mutation were identified in only one country, although a few exceptions to this were identified (Table 1; see Discussion). We tested the hypothesis that the strains with the same amino acid change were clonally related. This matter is important to address for public health and basic science reasons, because if these organisms are clonally related, it is unambiguous evidence that they can dissem-

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.



FIG 2 Aligned streptococcal PBP2X sequences. PBP2X of *S. pyogenes* strain MGAS5005 (GenBank accession number AAZ51984.1), *S. pneumoniae* (PBP reference sequence, GenBank accession number WP_050265832), and *S. agalactiae* (PBP reference sequence, GenBank accession number WP_134808185) were aligned by use of the ClustalW program. To facilitate comparisons, for each species, every 10th amino acid is in red. Amino acids of the consensus sequence are highlighted to indicate conserved domains, as indicated in the inset at the lower right. The three key conserved motifs (SXXK, SXN, and KSGT) of the transpeptidase are shown in red and bold below the aligned sequences.

inate successfully to new human hosts and cause infections. Phylogenetic analysis of whole-genome sequence data showed that, with one exception, strains of the same *emm* type with the identical amino acid replacement are closely related, likely as a consequence of descent from a common progenitor (Fig. 3). Exceptions were the 22 *emm28* strains with a Gly600Asp replacement (Fig. 3). These findings indicate multiple independent evolutionary origins of the Gly600Asp polymorphism, that is, multiple episodes of evolutionary convergence. We note that the single strain with the combined Phe599Tyr and Gly600Asp replacement was very closely related to two strains having only the single Gly600Asp change. This phylogenetic relationship suggests that the Phe599Tyr amino acid change was acquired (likely by selection) after the Gly600Asp change occurred in a progenitor. Consistent with this idea, the dual-amino-acid-replacement strain was isolated in 2006, years after the genetically related *emm28* strains with only the Gly600Asp replacement were initially found.

Association of PBP2X amino acid replacements with decreased susceptibility to beta-lactam antibiotics. We next tested the hypothesis that the PBP2X amino acid replacements are associated with decreased susceptibility to beta-lactam antibiotics. Strains were streaked onto tryptic soy agar plates supplemented with 6-ng/ml of penicillin G or 15-ng/ml ampicillin, and the plates were incubated overnight. These concentrations were previously determined to be minimally inhibitory for *S. pyogenes* (9, 10). Wild-type strains that lacked *pbp2x* mutations did not grow after overnight incubation in the presence of these beta-lactam antibiotics. In contrast, many strains with *pbp2x* nonsynonymous mutations grew well on both antibiotic-containing media, including organisms with the Gly288Ser, Met342Ile, Phe599Tyr plus Gly600Asp,

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

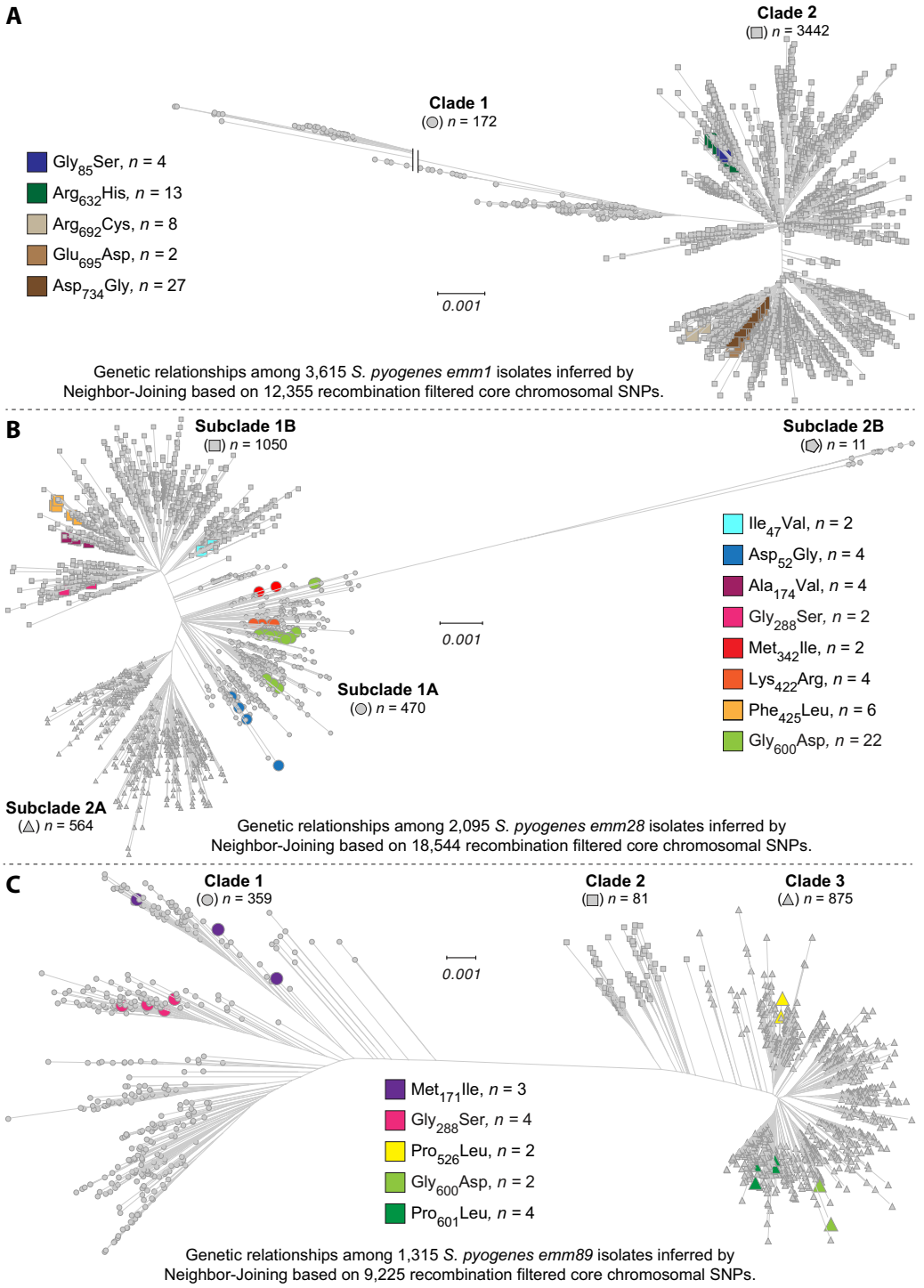


FIG 3 Genetic relationships among *S. pyogenes emm1*, *emm28*, and *emm89* clinical isolates. Phylogenies were inferred by neighbor joining based on core chromosomal SNPs. Isolates with nonsynonymous SNPs in *pbp2x* are colored according to the amino acid replacements in PBP2X, as shown in the keys. Clades of more closely related strains are shown by shapes (e.g., circles and squares), as indicated. (A) Relationships among 3,615 *emm1* strains; (B) relationships among 2,095 *emm28* strains; (C) relationships among 1,315 *emm89* strains.

Gly600Asp, and Pro601Leu amino acid replacements (Table 1; Fig. 1 and 3). Of note, in contrast to the five strains with the Pro601Leu replacement, the single *emm89* strain (MGAS27308) with the Pro601Ser change did not grow in the presence of either antibiotic under the plating conditions tested. Similarly, none of the 10 *emm1* strains

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

representing 8 different amino acid replacements grew under the antibiotic conditions tested. The data are consistent with the hypothesis of an association between some naturally occurring *pbp2x* mutations and decreased susceptibility to these beta-lactams in some genetic backgrounds. We next used Etest strips to determine the MICs for penicillin G and found that many strains with *pbp2x* mutations had decreased susceptibility to this agent, as tested in this fashion, whereas all 18 wild-type comparator strains lacking *pbp2x* mutations were fully susceptible (Table 1).

It is well-known that the same PBP2X amino acid replacement can confer divergent phenotypes of susceptibility to different beta-lactam antibiotics. Thus, we next performed MIC susceptibility testing with five additional beta-lactam antibiotics (ampicillin, cefotaxime, cefoxitin, ceftazidime, and meropenem) using the Etest gradient method. We found that, compared to the wild-type control strains, many PBP2X mutant strains had reduced susceptibility to one or more beta-lactam antibiotics (Table 1). The Etest MIC results for penicillin G and ampicillin were confirmed for some strains using broth microdilution and penicillin G and ampicillin agar (Table 1; Fig. 4; see also Fig. S1 in the supplemental material). Of note, strains with the Pro601Leu amino acid change, which occurred in both *emm28* and *emm89* strains, had the highest MIC measurements with all beta-lactam antibiotics tested except cefoxitin (Table 1). Specifically, the penicillin G MICs for strains with the Pro601Leu change ranged from 23 ng/ml to 32 ng/ml, approximately 4- to 5-fold higher than those for strains with the wild-type PBP2X (Table 1). To unambiguously demonstrate that the Pro601Leu PBP2X amino acid replacement was responsible for the altered MICs, we created an isogenic strain containing the wild-type *pbp2x* gene in place of the naturally occurring mutant *pbp2x* allele (encoding Pro601Leu). As expected, the isogenic Pro601 strain (i.e., the PBP2X consensus wild-type engineered derivative strain) was more susceptible to beta-lactam antibiotics than the naturally occurring parental strain with the Pro601Leu substitution (Fig. 4). Also, the strain with both the Phe599Tyr and Gly600Asp amino acid replacements had MIC measurements that were equal to or greater than those for strains with only the Gly600Asp change, suggesting that this dual amino acid replacement may have an additive effect on MICs.

Relative location of amino acid changes in the PBP2X three-dimensional structure. To assess the potential consequence of the identified amino acid replacements on PBP2X, we mapped the location of the changes on a crystal structure available for *S. pneumoniae* PBP2X (Fig. 5). The structure of this protein has been well studied by several investigators because of its importance in beta-lactam resistance in this common human pathogen (12–14, 16). The variant amino acids at positions 342, 599, 600, and 601 mapped to regions known to influence structure-function relationships (20, 21). This determination of influence was derived from overlaying PBP2X from *S. pneumoniae* with the clinically relevant and structurally similar *P. aeruginosa* PBP3 (PDB accession number 6UN3) and *S. aureus* PBP2a (PDB accession number 1VQQ, chain A). Recently, it was discovered that residues on the bottom of the α-8 helix of PBP3 are essential in forming an aromatic pocket (20) comprised of Tyr532 and Phe533. This aromatic pocket is key in binding and stabilizing the side chains of beta-lactam antibiotics. In PBP2X, a similar, conserved aromatic pocket is formed with the neighboring His594 and Tyr595 residues. The amino acid substitutions at residues 599, 600, and 601 that we observed in the clinical *S. pyogenes* isolates studied here are located directly above this aromatic pocket (Fig. 5). Substitutions at these positions may perturb binding interactions between the beta-lactam antibiotic and PBP2X and thereby decrease the acylation efficiency of the antibiotics, leading to reduced susceptibility. It is noteworthy that the Pro601Leu substitution was associated with decreased penicillin G susceptibility but that the Pro601Ser substitution was not. The serine side chain is relatively small and hydrophilic, whereas that of leucine is larger and hydrophobic, and both differ from proline, a secondary amino acid (i.e., an imino acid). Thus, both serine and leucine could potentially perturb the PBP2X structure and beta-lactam binding interactions but to a different extent, possibly leading to the differences in susceptibility observed.

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

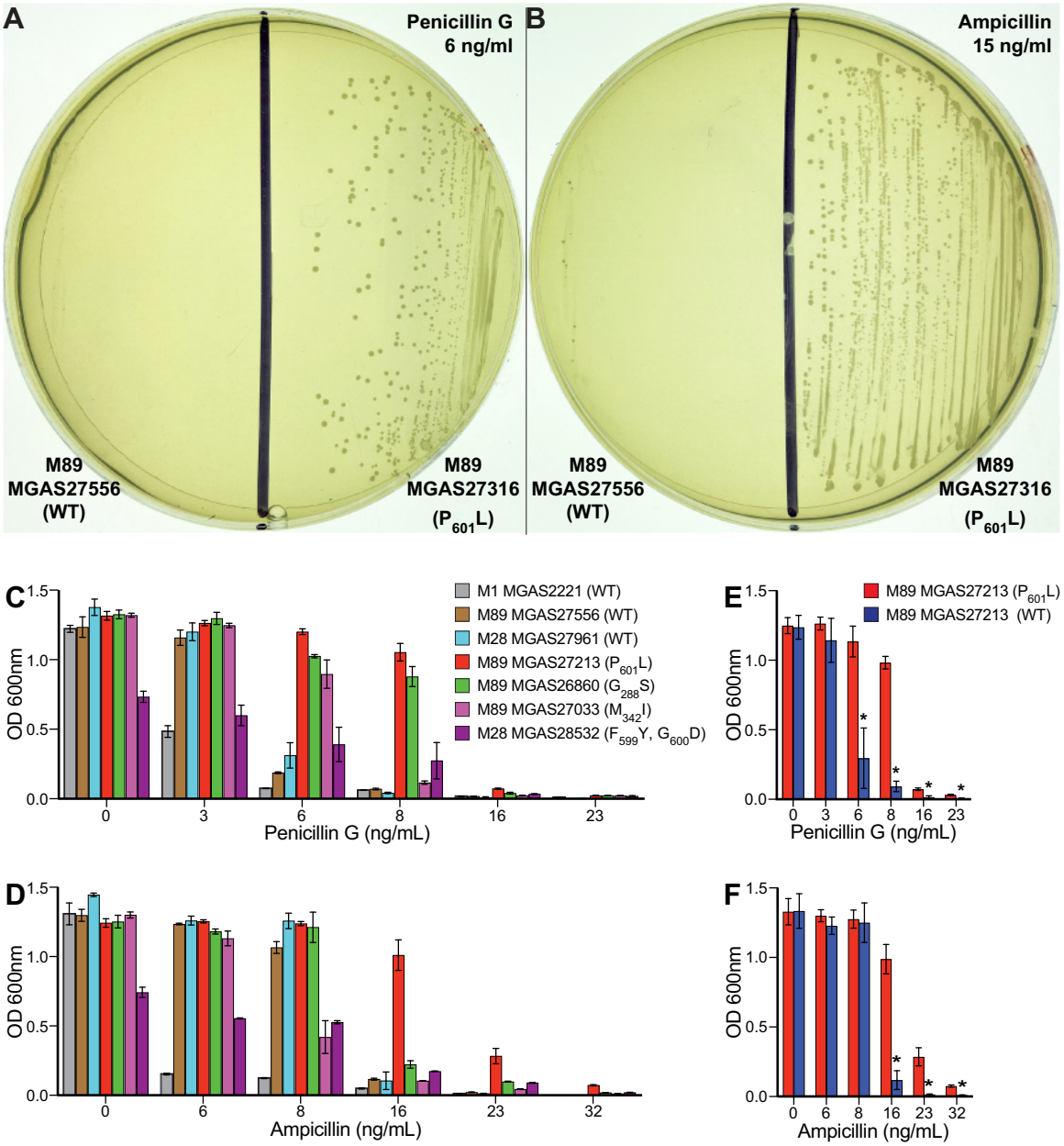


FIG 4 Beta-lactam antibiotic susceptibility assays. (A and B) Shown is the growth of *emm89* PBP2X wild-type strain MGAS27556 (left) in comparison with that of PBP2X Pro601Leu amino acid replacement strain MGAS27316 (right) on medium supplemented with 6-ng/ml penicillin G (A) or 15-ng/ml ampicillin (B). (C and D) Graphed is the MIC dilution growth of *emm1*, *emm28*, and *emm89* PBP2X wild-type and amino acid replacement variant strains in THY broth supplemented with penicillin G (C) or ampicillin (D). (E and F) Graphed is the growth of *emm89* PBP2X Pro601Leu replacement strain MGAS27213 and its isogenic PBP2X wild-type engineered derivative in THY broth supplemented with penicillin G (E) or ampicillin (F). All THY broth growth experiments were done in quadruplicate, and the results are given as the mean ± SD. *, significant differences in growth, as determined by Student's *t* test at a *P* value of <0.05. OD_{600}, optical density at 600 nm.

Similarly, the Met342Ile substitution is noteworthy because residue 342 is located directly within the active-site pocket of PBP2X near the catalytic Ser337. In PBP2X, a methionine cluster is conserved within the active site (15). Thus, the Met342Ile substitution present in the clinical *S. pyogenes* isolate may disrupt the conserved methionine cluster and thereby perturb binding and acylation of the enzyme by beta-lactam antibiotics. Decreased acylation efficiency, in turn, would explain the reduced beta-lactam susceptibility associated with clinical *S. pyogenes* isolates containing amino acid variants at position 342. In contrast, the Gly288Ser substitution is not located in a

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

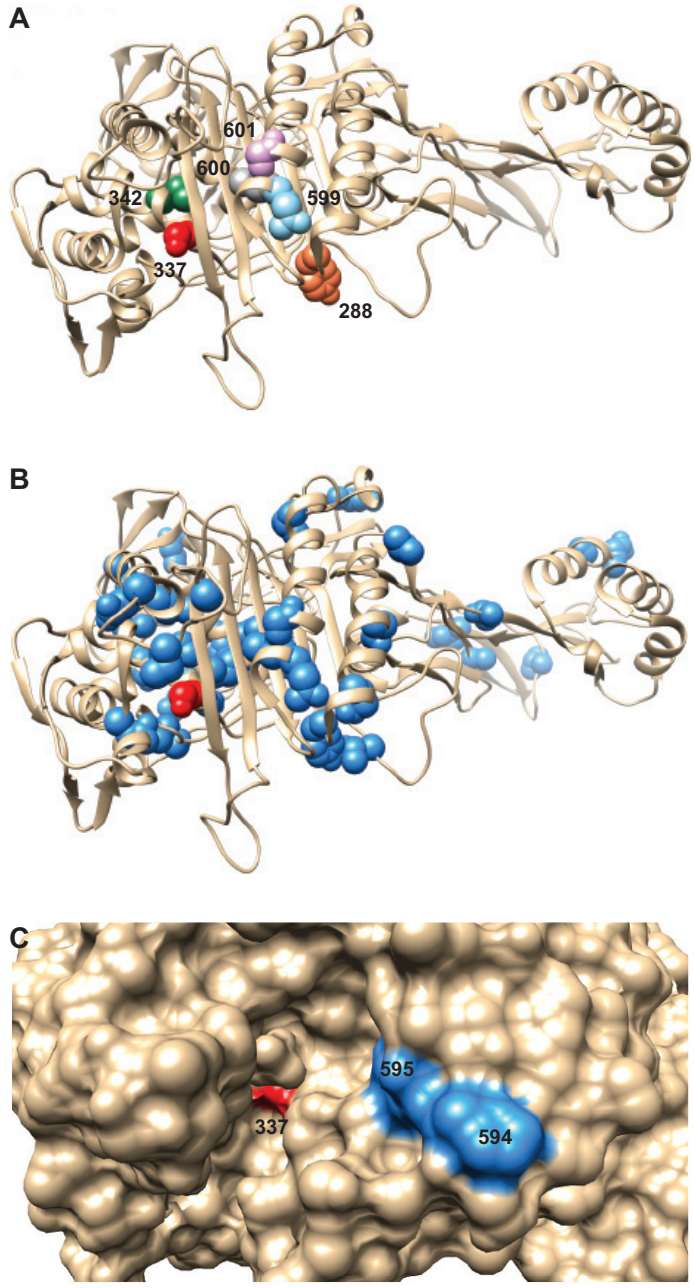


FIG 5 Location of *S. pyogenes* PBP2X substitutions relative to the X-ray crystallography structure of PBP2X from *S. pneumoniae* (PDB accession number 1RP5, chain A). (A) Variant sites influencing structure-function. Illustrated as spheres on the *S. pneumoniae* PBP2X ribbon diagram are the key amino acid replacement sites associated with reduced beta-lactam susceptibility from the *S. pyogenes* clinical isolates, with the relative amino acid positions being labeled. Shown in red is the transpeptidase catalytic serine residue (*S. pneumoniae* residue 337 = *S. pyogenes* residue 340). The amino acids depicted are those of the *S. pneumoniae* PBP2X. (B) All variant sites. The relative positions of all 36 amino acid replacement sites observed among the 7,025 sequenced strains are shown as blue spheres, with the catalytic serine being shown in red. (C) Aromatic pocket. Illustrated is a surface representation showing the aromatic pocket proposed to be involved in binding and stabilizing beta-lactam side chains. The His594 and Tyr595 residues lining the pocket are shown in blue.

position to directly impact substrate binding or acylation (Fig. 5). Indicative of this, the C_α of residue 288 is 17.5 Å from the O_γ of the catalytic Ser337 in the *S. pneumoniae* PBP2X structure. Regardless, the substitution could detrimentally impact enzyme dynamics or stability to alter function. A more definitive conclusion on how the Gly288Ser substitution alters enzyme structure and function awaits further work.

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

DISCUSSION

Here we report on 137 strains of *S. pyogenes* from intercontinental sources that have 37 amino acid replacements at 36 sites in the PBP2X protein, some of which correlate with decreased susceptibility to beta-lactam antibiotics under the conditions tested. Importantly, none of the mutations that we identified resulted in resistance *in vitro* to any of the six beta-lactams studied (as defined by CLSI), and none approached the level for ampicillin or cefotaxime MICs described for the PBP2X substitution Thr553Lys. This substitution evidently conferred an MIC at the CLSI-determined breakpoint for nonsusceptibility to ampicillin. However, isogenic mutant strains were not constructed to prove that the mutant allele of *pbp2x* was solely responsible for the altered MIC value. This is an important point, because the two strains described by Vannice et al. (3) also contained a Ser79Phe amino acid replacement in the topoisomerase ParC. Substitutions in ParC can confer resistance to fluoroquinolone antibiotics. In principle, substitutions in ParC could produce a slowed growth phenotype potentially contributing to the altered beta-lactam MICs observed. As described by Vannice et al. (3), all five *emm43.4* strains that they analyzed also had a Thr236Ala amino acid replacement in a gene annotated "glycoside hydrolase family 25." The only known enzymatic activity of the family 25 glycoside hydrolases is that of a lysozyme muramidase. Members of this protein family participate in peptidoglycan remodeling, and thus, in principle, the Thr236Ala substitution might also contribute to the altered beta-lactam MIC. Clearly, much more work using isogenic mutant strains is required to deconvolute the role of specific amino acid replacements in these proteins in the observed altered MICs.

Our research was stimulated by the recent description of two clonally related type *emm43.4 S. pyogenes* strains with the same Thr553Lys amino acid replacement in PBP2X associated with altered MICs of beta-lactam antibiotics (3). We note that the Thr553Lys change likely reflects very recent antimicrobial selection, in view of the described course of treatment of the two host patients. Our work was made possible in part by the availability of 7,025 genome sequences from geographically dispersed strains of types *emm1*, *emm28*, and *emm89* that we previously generated for molecular pathogenesis, population genomic, and epidemiological purposes. This unique resource permitted us to rapidly identify strains with mutations in the *pbp2x* gene by bioinformatic methods and subsequently assess the beta-lactam susceptibility phenotypes by standard clinical microbiology methods. Our findings indicate that the decreased beta-lactam susceptibility associated with some PBP2X amino acid polymorphisms in this pathogen is geographically widespread and has arisen multiple times independently over many years in contemporary epidemic clones of serotype *emm1*, *emm28*, and *emm89* GAS (Fig. 2). The data support the interpretation that the nonsynonymous mutations have been selected by exposure to beta-lactam antibiotics used during treatment of infections caused by *S. pyogenes*. However, inasmuch as *S. pyogenes* can be carried asymptomatically in the upper respiratory tract or other anatomic site, it is also possible that selection occurred during antibiotic treatment of an asymptomatic carrier for an infection caused by another organism. The potential for reduced beta-lactam susceptibility to confer an advantage during human infections (either invasive or nonsystemic infection, such as pharyngitis or localized skin and soft tissue infections) or asymptomatic carriage remains untested. More study is required to address these important issues.

Examples of convergent evolution. Several examples of convergent evolution to decreased beta-lactam susceptibility were identified in this analysis. Four instances of the presence of the same otherwise rare single amino acid polymorphism in strains of different *emm* types were found. We identified strains of *emm28* and *emm89* with each of the following amino acid replacements: Gly288Ser, Met342Ile, Gly600Asp, or Pro601Leu. Given that strains of *emm28* and *emm89* are very distantly related genetically, the only reasonable interpretation is that these polymorphisms arose independently as a consequence of convergent evolution, presumably due to selection following exposure to a beta-lactam antibiotic. Similarly, the occurrence of the Gly600Asp

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

replacement in some *emm28* strains that have not shared a recent common ancestor serves as another clear example of convergent evolution in *pbp2x*. As further evidence of convergent evolution, Chochua et al. also found the Pro601Leu amino acid change in multiple unrelated GAS lineages, including *emm4* and *emm75* isolates containing it as a single amino acid replacement and *emm87* and *emm89* isolates containing it in combination with a second substitution (4).

Why is there an apparent difference between *emm1* strains and *emm28* and *emm89* strains? The majority of strains that we identified with *pbp2x* nonsynonymous mutations were either type *emm28* or type *emm89* subclade 3 organisms. In addition, the linear location of PBP2X amino acid replacements in the *emm1* strains differed from that in the *emm28* and *emm89* strains. We believe that there are several factors that may contribute to these differences. First, it is important to note that essentially all *emm28* organisms do not produce hyaluronic acid capsule, as a consequence of having an insertion of an adenine nucleotide after nucleotide 219 in an A-T-rich region (7). This single nucleotide insertion severely truncates *hasA*, whose gene product is required for capsule biosynthesis. Similarly, *emm89* organisms of subclade 3 fail to make hyaluronic acid capsule because they lack the *hasABC* operon, required for capsule biosynthesis (5, 22). The hyaluronic capsule, among other interactions with the host, contributes to the capacity of *S. pyogenes* to resist phagocytosis. It is possible that a relationship exists between the inability of an *S. pyogenes* strain to produce hyaluronic acid capsule and the likelihood of generating a strain that is extant and that has PBP2X amino acid changes that result in decreased beta-lactam susceptibility. Second, it is possible that the *in vivo* regulation and/or expression of *pbp2x* differs between strains of distinct clonal backgrounds. A third possibility is that, for unknown reasons, *emm1* strains with PBP2X amino acid replacements are simply less fit *in vivo* than *emm28* and *emm89* strains. A fourth possibility is that the *in vivo* topology of PBP2X differs between *emm1* strains and *emm28* and *emm89* strains, perhaps due to interaction with other currently unknown proteins. However, all of these ideas are speculative, and more study is required to address these important observations. Finally, we note that the possibilities described above are not mutually exclusive.

Relationship of our findings to those reported for other pathogenic beta-hemolytic streptococci. Strains of *Streptococcus agalactiae* and *Streptococcus dysgalactiae* subsp. *equisimilis* with decreased susceptibility to beta-lactam antibiotics have been reported (23–39). In the case of *S. dysgalactiae* subsp. *equisimilis*, four isolates cultured from the blood of three epidemiologically associated patients were reported to be resistant to penicillin and oxacillin (25). Whole-genome sequencing identified nonsynonymous mutations in PBP2X that were thought to be causally involved in the resistance phenotype. In particular, the investigators identified the occurrence of Thr341Pro and Gln555Glu amino acid replacements and noted that these two changes are located close to positions 337, 547, and 557, which are among the more prevalently found variant sites reported for penicillin-resistant *S. pneumoniae*. As described above, we identified decreased susceptibility in *S. pyogenes* strains with an amino acid change at position 342 (Met342Ile), and the altered amino acid reported by Vannice et al. was Thr553Lys (3).

Public health implications. Our population genomic analysis indicates that *S. pyogenes* strains with nonsynonymous mutations in *pbp2x* are not exceedingly rare in the *emm1*, *emm28*, and *emm89* organisms that we studied, being present in approximately 2% of the collection of 7,025 isolates studied. Although this relatively low frequency is fortunate, two facts give us pause. First, the great majority of the isolates that we previously characterized by whole-genome sequencing were cultured from patients with invasive episodes. Given the relative lack of strains causing pharyngitis in this sample, coupled with the well-documented treatment failures occurring among individuals with culture-positive *S. pyogenes* pharyngitis (40–48), it is possible that analysis of large samples of strains from pharyngitis patients will identify a different percentage of organisms with *pbp2x* mutations associated with altered susceptibility to beta-lactam antibiotics. Second, we identified

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

strains with PBP2X amino acid changes that are clearly clonally related based on phylogenetic analysis of whole-genome data. For example, we recovered clonally related type *emm28* organisms with the Gly600Asp replacement from patients in Canada and five different states in the United States, indicating that they can successfully disseminate over geographic distances and cause infections. The same is true for the 27 clonally related strains of type *emm1* with the Asp734Gly change, organisms causing invasive infections in Denmark, Sweden, and Iceland between 2002 and 2007 (Table 1). Taken together, our findings stress the importance of renewed efforts to monitor antimicrobial susceptibility rates and values in this pathogen on an ongoing basis, the need to formulate an efficacious human vaccine against *S. pyogenes*, and the need for expanded vaccine efforts, as noted by many (49–51). Importantly, these needs also were highlighted in the report of a symposium held more than 2 decades ago dedicated to the topic of a lack of penicillin resistance in *S. pyogenes* (1).

What may the future hold? Although some favor the idea that we are not at the beginning of the end of the universal susceptibility of *S. pyogenes* to beta-lactam antibiotics (52), we believe that there are multiple reasons to be less sanguine in the long term. First, our data show that several distinct *pbp2x* mutations associated with decreased susceptibility occur in *S. pyogenes* strains of multiple *emm* types. Second, in contrast to the otherwise rare *emm43.4* strains reported by Vannice et al. (3), we identified PBP2X amino acid replacements in strains of types *emm1*, *emm28*, and *emm89*, which, in the aggregate, are common causes of *S. pyogenes* pharyngitis and invasive episodes in many countries. Third, some of the organisms with amino acid changes are clonally related and have been recovered in multiple geographic locations, in some cases many years apart. Thus, if strains with PBP2X amino acid replacements have decreased fitness, at least in some cases the deficit is not sufficient to prohibit the successful dissemination of some clonal progeny to new hosts and the capacity to cause serious human invasive infections. Fourth, the exchange of genetic material between *S. pyogenes* strains can produce progeny with altered phenotypes, such as enhanced virulence and increased antimicrobial agent resistance. Thus, in principle, there is a risk of gene flow of a mutant *pbp2x* gene to a susceptible strain, a process that could accelerate the spread of decreased susceptibility to beta-lactams or frank resistance in this global human pathogen.

Concluding comment. To summarize, we used our library of 7,025 *S. pyogenes* genome sequences from strains of types *emm1*, *emm28*, and *emm89* to identify amino acid-altering mutations in *pbp2x*. Some of the strains with amino acid replacements in PBP2X had decreased susceptibility under the laboratory conditions tested to some beta-lactam antibiotics, including the commonly used penicillin G. Although many *pbp2x* mutations occurred in only one or two strains, we found that some PBP2X amino acid replacements were present in multiple clonally related strains causing infections many years apart. Decreased susceptibility to beta-lactams in *S. pyogenes* is geographically widespread and exists in strains of numerically common *emm* gene subtypes. We recommend that increased basic science and translational research attention be applied to this potentially severe public health threat. For example, the availability of an efficacious human vaccine against *S. pyogenes* pharyngitis would significantly decrease the use of beta-lactam antibiotic agents globally. We believe that for diagnostic laboratories not currently routinely performing beta-lactam susceptibility testing, it is reasonable to consider doing so, perhaps by measuring the penicillin MIC.

SUPPLEMENTAL MATERIAL

Supplemental material is available online only.

SUPPLEMENTAL FILE 1, PDF file, 0.2 MB.

ACKNOWLEDGMENTS

This study was supported in part by the Fondren Foundation, the Houston Methodist Hospital and Research Institute, and National Institutes of Health grants AI139369 and AI146771 (to J.M.M.) and AI32956 (to T.P.).

We thank the dedicated staff of the Houston Methodist Hospital and Research

Institute Clinical Microbiology Laboratory (including Oluwatobi Adelanwa, Edevelia Cornelius, Lily Guevara, and Patricia L. Cernoch) for assistance in performing the Etest MIC determinations; Concepcion C. Cantu, Matthew Ojeda Saavedra, Layne Pruitt, and Prasanti Yerramilli for technical assistance; Jari Jalava, Carita Savolainen-Kopra, and Outi Lyytikäinen for expert opinion; Kati Räisänen and Tuula Siljander for *emm* typing of strains; and the Finnish clinical microbiology laboratories for sending the laboratory notification and strains to THL.

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

REFERENCES

1. Horn DL, Zabriskie JB, Austrian R, Cleary PP, Ferretti JJ, Fischetti VA, Gotschlich E, Kaplan EL, McCarty M, Opal SM, Roberts RB, Tomasz A, Wachtfogel Y. 1998. Why have group A streptococci remained susceptible to penicillin? Report on a symposium. Clin Infect Dis 26:1341–1345. https://doi.org/10.1086/516375.
2. Ralph AP, Carapetis JR. 2013. Group A streptococcal diseases and their global burden. Curr Top Microbiol Immunol 368:1–27. https://doi.org/10.1007/82_2012_280.
3. Vannice K, Ricaldi J, Nanduri S, Fang FC, Lynch J, Bryson-Cahn C, Wright T, Duchin J, Kay M, Chochua S, Van Beneden C, Beall B. 20 October 2019. Streptococcus pyogenes pbp2x mutation confers reduced susceptibility to beta-lactam antibiotics. Clin Infect Dis https://doi.org/10.1093/cid/ciz1000.
4. Chochua S, Metcalf BJ, Li Z, Rivers J, Mathis S, Jackson D, Gertz RE, Jr, Srinivasan V, Lynfield R, Van Beneden C, McGee L, Beall B. 2017. Population and whole genome sequence based characterization of invasive group A streptococci recovered in the United States during 2015. mBio 8:e01422-17. https://doi.org/10.1128/mBio.01422-17.
5. Beres SB, Kachroo P, Nasser W, Olsen RJ, Zhu L, Flores AR, de la Riva I, Paez-Mayorga J, Jimenez FE, Cantu C, Vuopio J, Jalava J, Kristinsson KG, Gottfredsson M, Corander J, Fittipaldi N, Di Luca MC, Petrelli D, Vitali LA, Raiford A, Jenkins L, Musser JM. 2016. Transcriptome remodeling contributes to epidemic disease caused by the human pathogen Streptococcus pyogenes. mBio 7:e00403-16. https://doi.org/10.1128/mBio.00403-16.
6. Beres SB, Olsen RJ, Ojeda Saavedra M, Ure R, Reynolds A, Lindsay DSJ, Smith AJ, Musser JM. 2017. Genome sequence analysis of emm89 Streptococcus pyogenes strains causing infections in Scotland, 2010–2016. J Med Microbiol 66:1765–1773. https://doi.org/10.1099/jmm.0.000622.
7. Kachroo P, Eraso JM, Beres SB, Olsen RJ, Zhu L, Nasser W, Bernard PE, Cantu CC, Saavedra MO, Arredondo MJ, Strope B, Do H, Kumaraswami M, Vuopio J, Grondahl-Yli-Hannuksela K, Kristinsson KG, Gottfredsson M, Pesonen M, Pensar J, Davenport ER, Clark AG, Corander J, Caugant DA, Gaini S, Magnussen MD, Kubiak SL, Nguyen HAT, Long SW, Porter AR, DeLeo FR, Musser JM. 2019. Integrated analysis of population genomics, transcriptomics and virulence provides novel insights into Streptococcus pyogenes pathogenesis. Nat Genet 51:548–559. https://doi.org/10.1038/s41588-018-0343-1.
8. Nasser W, Beres SB, Olsen RJ, Dean MA, Rice KA, Long SW, Kristinsson KG, Gottfredsson M, Vuopio J, Raisanen K, Caugant DA, Steinbakk M, Low DE, McGeer A, Darenberg J, Henriques-Normark B, Van Beneden CA, Hoffmann S, Musser JM. 2014. Evolutionary pathway to increased virulence and epidemic group A Streptococcus disease derived from 3,615 genome sequences. Proc Natl Acad Sci U S A 111:E1768–E1776. https://doi.org/10.1073/pnas.1403138111.
9. Gutmann L, Tomasz A. 1982. Penicillin-resistant and penicillin-tolerant mutants of group A streptococci. Antimicrob Agents Chemother 22:128–136. https://doi.org/10.1128/aac.22.1.128.
10. Sakata H. 2013. Susceptibility and emm type of Streptococcus pyogenes isolated from children with severe infection. J Infect Chemother 19:1042–1046. https://doi.org/10.1007/s10156-013-0617-6.
11. Zhu L, Olsen RJ, Nasser W, Beres SB, Vuopio J, Kristinsson KG, Gottfredsson M, Porter AR, DeLeo FR, Musser JM. 2015. A molecular trigger for intercontinental epidemics of group A Streptococcus. J Clin Invest 125:3545–3559. https://doi.org/10.1172/JCI82478.
12. Gordon E, Mouz N, Duee E, Dideberg O. 2000. The crystal structure of the penicillin-binding protein 2x from Streptococcus pneumoniae and its acyl-enzyme form: implication in drug resistance. J Mol Biol 299:477–485. https://doi.org/10.1006/jmbi.2000.3740.
13. Mouz N, Di Guilmi AM, Gordon E, Hakenbeck R, Dideberg O, Vernet T. 1999. Mutations in the active site of penicillin-binding protein PBP2x from Streptococcus pneumoniae. Role in the specificity for beta-lactam antibiotics. J Biol Chem 274:19175–19180. https://doi.org/10.1074/jbc.274.27.19175.
14. Mouz N, Gordon E, Di Guilmi AM, Petit I, Petillot Y, Dupont Y, Hakenbeck R, Vernet T, Dideberg O. 1998. Identification of a structural determinant for resistance to beta-lactam antibiotics in Gram-positive bacteria. Proc Natl Acad Sci U S A 95:13403–13406. https://doi.org/10.1073/pnas.95.23.13403.
15. Pares S, Mouz N, Petillot Y, Hakenbeck R, Dideberg O. 1996. X-ray structure of Streptococcus pneumoniae PBP2x, a primary penicillin target enzyme. Nat Struct Biol 3:284–289. https://doi.org/10.1038/nsb0396-284.
16. Pernot L, Chesnel L, Le Gouellec A, Croize J, Vernet T, Dideberg O, Dessen A. 2004. A PBP2x from a clinical isolate of Streptococcus pneumoniae exhibits an alternative mechanism for reduction of susceptibility to beta-lactam antibiotics. J Biol Chem 279:16463–16470. https://doi.org/10.1074/jbc.M313492200.
17. Pettersen EF, Goddard TD, Huang CC, Couch GS, Greenblatt DM, Meng EC, Ferrin TE. 2004. UCSF Chimera—a visualization system for exploratory research and analysis. J Comput Chem 25:1605–1612. https://doi.org/10.1002/jcc.20084.
18. Le Breton Y, Belew AT, Valdes KM, Islam E, Curry P, Tettelin H, Shirtliff ME, El-Sayed NM, McIver KS. 2015. Essential genes in the core genome of the human pathogen Streptococcus pyogenes. Sci Rep 5:9838. https://doi.org/10.1038/srep09838.
19. Zhu L, Olsen RJ, Beres SB, Eraso JM, Saavedra MO, Kubiak SL, Cantu CC, Jenkins L, Charbonneau ARL, Waller AS, Musser JM. 2019. Gene fitness landscape of group A streptococcus during necrotizing myositis. J Clin Invest 129:887–901. https://doi.org/10.1172/JCI124994.
20. Sacco MD, Kroeck KG, Kemp MT, Zhang X, Andrews LD, Chen Y. 2019. Influence of the alpha-methoxy group on the reaction of temocillin with Pseudomonas aeruginosa PBP3 and CTX-M-14 beta-lactamase. Antimicrob Agents Chemother 64:e01473-19. https://doi.org/10.1128/AAC.01473-19.
21. Zapun A, Contreras-Martel C, Vernet T. 2008. Penicillin-binding proteins and beta-lactam resistance. FEMS Microbiol Rev 32:361–385. https://doi.org/10.1111/j.1574-6976.2007.00095.x.
22. Turner CE, Abbott J, Lamagni T, Holden MT, David S, Jones MD, Game L, Efstratiou A, Sriskandan S. 2015. Emergence of a new highly successful acapsular group A streptococcus clade of genotype emm89 in the United Kingdom. mBio 6:e00622-15. https://doi.org/10.1128/mBio.00622-15.
23. Banno H, Kimura K, Seki T, Jin W, Wachino JI, Yamada K, Nagano N, Arakawa Y. 2018. High isolation rate and multidrug resistance tendency of penicillin-susceptible group B Streptococcus with reduced ceftibuten susceptibility in Japan. Eur J Clin Microbiol Infect Dis 37:1511–1519. https://doi.org/10.1007/s10096-018-3278-7.
24. Dahesh S, Hensler ME, Van Sorge NM, Gertz RE, Jr, Schrag S, Nizet V, Beall BW. 2008. Point mutation in the group B streptococcal pbp2x gene conferring decreased susceptibility to beta-lactam antibiotics. Antimicrob Agents Chemother 52:2915–2918. https://doi.org/10.1128/AAC.00461-08.
25. Fuursted K, Stegger M, Hoffmann S, Lambertsen L, Andersen PS, Deleuran M, Thomsen MK. 2016. Description and characterization of a penicillin-resistant Streptococcus dysgalactiae subsp. equisimilis clone isolated from blood in three epidemiologically linked patients. J Antimicrob Chemother 71:3376–3380. https://doi.org/10.1093/jac/dkw320.
26. Gaudreau C, Lecours R, Ismail J, Gagnon S, Jette L, Roger M. 2010. Prosthetic hip joint infection with a Streptococcus agalactiae isolate not susceptible to penicillin G and ceftriaxone. J Antimicrob Chemother 65:594–595. https://doi.org/10.1093/jac/dkp458.
27. Kamiya C, Kimura K, Doyama Y, Miyazaki A, Morimoto M, Banno H,

Downloaded from https://journals.asm.org/journal/jcm on 09 August 2026 by 2401:4900:c918:ddd9:4dac:92e8:bc74:f4f1.

Nagano N, Jin W, Wachino J, Yamada K, Arakawa Y. 2015. Ceftibuten-containing agar plate for detecting group B streptococci with reduced penicillin susceptibility (PRGBS). Diagn Microbiol Infect Dis 82:269–273. https://doi.org/10.1016/j.diagmicrobio.2015.04.010.

28. Kimura K, Nagano N, Arakawa Y. 2015. Classification of group B streptococci with reduced beta-lactam susceptibility (GBS-RBS) based on the amino acid substitutions in PBPs. J Antimicrob Chemother 70:1601–1603. https://doi.org/10.1093/jac/dkv022.
29. Kimura K, Nagano N, Nagano Y, Wachino J, Suzuki S, Shibayama K, Arakawa Y. 2011. Predominance of sequence type 1 group with serotype VI among group B streptococci with reduced penicillin susceptibility identified in Japan. J Antimicrob Chemother 66:2460–2464. https://doi.org/10.1093/jac/dkr352.
30. Kimura K, Suzuki S, Wachino J, Kurokawa H, Yamane K, Shibata N, Nagano N, Kato H, Shibayama K, Arakawa Y. 2008. First molecular characterization of group B streptococci with reduced penicillin susceptibility. Antimicrob Agents Chemother 52:2890–2897. https://doi.org/10.1128/AAC.00185-08.
31. Kimura K, Wachino J, Kurokawa H, Matsui M, Suzuki S, Yamane K, Nagano N, Shibayama K, Arakawa Y. 2013. High cephalosporin resistance due to amino acid substitutions in PBP1A and PBP2X in a clinical isolate of group B Streptococcus. J Antimicrob Chemother 68:1533–1536. https://doi.org/10.1093/jac/dkt060.
32. Koide S, Hayashi W, Taniguchi Y, Tanaka H, Kimura K, Nagano Y, Arakawa Y, Nagano N. 2019. Potential effect of selective pressure with different beta-lactam molecules on the emergence of reduced susceptibility to beta-lactams in group B streptococci. Microbiol Immunol 63:65–76. https://doi.org/10.1111/1348-0421.12667.
33. Longtin J, Vermeiren C, Shahinas D, Tamber GS, McGeer A, Low DE, Katz K, Pillai DR. 2011. Novel mutations in a patient isolate of Streptococcus agalactiae with reduced penicillin susceptibility emerging after long-term oral suppressive therapy. Antimicrob Agents Chemother 55:2983–2985. https://doi.org/10.1128/AAC.01243-10.
34. Metcalf BJ, Chochua S, Gertz RE, Jr, Hawkins PA, Ricaldi J, Li Z, Walker H, Tran T, Rivers J, Mathis S, Jackson D, Glennen A, Lynfield R, McGee L, Beall B, Active Bacterial Core Surveillance Team. 2017. Short-read whole genome sequencing for determination of antimicrobial resistance mechanisms and capsular serotypes of current invasive Streptococcus agalactiae recovered in the USA. Clin Microbiol Infect 23:574.e7–574.e14. https://doi.org/10.1016/j.cmi.2017.02.021.
35. Nagano N, Kimura K, Nagano Y, Yakumaru H, Arakawa Y. 2009. Molecular characterization of group B streptococci with reduced penicillin susceptibility recurrently isolated from a sacral decubitus ulcer. J Antimicrob Chemother 64:1326–1328. https://doi.org/10.1093/jac/dkp374.
36. Nagano N, Nagano Y, Kimura K, Tamai K, Yanagisawa H, Arakawa Y. 2008. Genetic heterogeneity in pbp genes among clinically isolated group B streptococci with reduced penicillin susceptibility. Antimicrob Agents Chemother 52:4258–4267. https://doi.org/10.1128/AAC.00596-08.
37. Nagano N, Nagano Y, Toyama M, Kimura K, Shibayama K, Arakawa Y. 2014. Penicillin-susceptible group B streptococcal clinical isolates with reduced cephalosporin susceptibility. J Clin Microbiol 52:3406–3410. https://doi.org/10.1128/JCM.01291-14.
38. Nagano N, Nagano Y, Toyama M, Kimura K, Tamura T, Shibayama K, Arakawa Y. 2012. Nosocomial spread of multidrug-resistant group B streptococci with reduced penicillin susceptibility belonging to clonal complex 1. J Antimicrob Chemother 67:849–856. https://doi.org/10.1093/jac/dkr546.
39. Suzuki T, Kimura K, Suzuki H, Banno H, Jin W, Wachino J, Yamada K, Arakawa Y. 2015. Have group A streptococci with reduced penicillin susceptibility emerged? J Antimicrob Chemother 70:1258–1259. https://doi.org/10.1093/jac/dku492.
40. Brook I. 2007. Overcoming penicillin failures in the treatment of group A streptococcal pharyngo-tonsillitis. Int J Pediatr Otorhinolaryngol 71:1501–1508. https://doi.org/10.1016/j.ijporl.2007.06.006.
41. Brook I. 2017. Treatment challenges of group A beta-hemolytic streptococcal pharyngo-tonsillitis. Int Arch Otorhinolaryngol 21:286–296. https://doi.org/10.1055/s-0036-1584294.
42. Gastanaduy AS, Kaplan EL, Huwe BB, McKay C, Wannamaker LW. 1980. Failure of penicillin to eradicate group A streptococci during an outbreak of pharyngitis. Lancet ii:498–502. https://doi.org/10.1016/S0140-6736(80)91832-2.
43. Kim KS, Kaplan EL. 1985. Association of penicillin tolerance with failure to eradicate group A streptococci from patients with pharyngitis. J Pediatr 107:681–684. https://doi.org/10.1016/S0022-3476(85)80392-9.
44. Musser JM, Gray BM, Schlievert PM, Pichichero ME. 1992. Streptococcus pyogenes pharyngitis: characterization of strains by multilocus enzyme genotype, M and T protein serotype, and pyrogenic exotoxin gene probing. J Clin Microbiol 30:600–603. https://doi.org/10.1128/JCM.30.3.600-603.1992.
45. Pichichero ME. 1992. Explanations and therapies for penicillin failure in streptococcal pharyngitis. Clin Pediatr (Phila) 31:642–649. https://doi.org/10.1177/000992289203101101.
46. Pichichero ME, Casey JR. 2007. Systematic review of factors contributing to penicillin treatment failure in Streptococcus pyogenes pharyngitis. Otolaryngol Head Neck Surg 137:851–857. https://doi.org/10.1016/j.otohns.2007.07.033.
47. Pichichero ME, Gooch WM, Rodriguez W, Blumer JL, Aronoff SC, Jacobs RF, Musser JM. 1994. Effective short-course treatment of acute group A beta-hemolytic streptococcal tonsillopharyngitis. Ten days of penicillin V vs 5 days or 10 days of cefpodoxime therapy in children. Arch Pediatr Adolesc Med 148:1053–1060. https://doi.org/10.1001/archpedi.1994.02170100051010.
48. Pichichero ME, Green JL, Francis AB, Marsocci SM, Murphy AM, Hoeger W, Noriega C, Sorrento A, Gootnick J. 1998. Recurrent group A streptococcal tonsillopharyngitis. Pediatr Infect Dis J 17:809–815. https://doi.org/10.1097/00006454-199809000-00012.
49. Dale JB, Batzloff MR, Cleary PP, Courtney HS, Good MF, Grandi G, Halperin S, Margarit IY, McNeil S, Pandey M, Smeesters PR, Steer AC. 2016. Current approaches to group A streptococcal vaccine development. *In* Ferretti JJ, Stevens DL, Fischetti VA (ed), Streptococcus pyogenes: basic biology to clinical manifestations. University of Oklahoma Health Sciences Center, Oklahoma City, OK.
50. Steer AC, Carapetis JR, Dale JB, Fraser JD, Good MF, Guilherme L, Moreland NJ, Mulholland EK, Schodel F, Smeesters PR. 2016. Status of research and development of vaccines for Streptococcus pyogenes. Vaccine 34:2953–2958. https://doi.org/10.1016/j.vaccine.2016.03.073.
51. Vekemans J, Gouvea-Reis F, Kim JH, Excler J-L, Smeesters PR, O'Brien KL, Van Beneden CA, Steer AC, Carapetis JR, Kaslow DC. 2019. The path to group A streptococcus vaccines: World Health Organization research and development technology roadmap and preferred product characteristics. Clin Infect Dis 69:877–883. https://doi.org/10.1093/cid/ciy1143.
52. Hanage WP, Shelburne SA. 20 October 2019. Streptococcus pyogenes with reduced susceptibility to beta-lactams: how big an alarm bell? Clin Infect Dis https://doi.org/10.1093/cid/ciz1006.